

Automated Segmentation for Archaeological Rock Art: A Comparative Pipeline Study

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Abstract

Segmentation of rock art at vast heritage sites is a tedious manual activity, and field archaeologists now have a growing demand for automatic algorithms that can extract painted or engraved figures against a highly textured stone background. In this study, we proposed a rock art segmentation pipeline based on contrast-limited adaptive histogram equalization (CLAHE) and bilateral filtering, followed by the extraction of artistic motifs against the background rock, which is the actual segmentation problem we solved. We evaluated six configurations to assess the impact of image enhancements on the final output quality, including standard Otsu threshold contour, CLAHE enhanced Canny, marker-controlled watershed algorithm, pre-trained COCO Mask R-CNN, trained UNet, and trained UNet with post-processing. The mean Intersection over Union (IoU) was our comparative metric throughout this study. Among the important results obtained was the fact that combining CLAHE with a standard Canny detector reduces the performance of the latter, dropping its IoU value from 23.7% to 18.2%, implying that contrast-enhanced images do not add much value in the case of overly rigid segmentation algorithms.

Keywords: Image Segmentation, Rock Art · Digital Heritage, UNet, Computer Vision, Deep Learning.

1 Introduction

Pictographs painted on rocks and petroglyphs inscribed on rocks are still found at heritage sites worldwide, providing a direct glimpse into the most primitive forms of human expression. In places like India's Bhimbetka rock shelters and Saru-Maruv Caves, where hundreds of individual figures still await documentation, it takes months of field research to document all these figures

manually, making the use of computational techniques necessary because of the large number of undocumented artworks.

Three factors complicate the technical aspects of this issue. First, motifs tend to be worn out and can be overlain with mineral crusts and lichens and painted on rocky surfaces that have natural crackling and striations that look like brushstrokes that paint the motifs. Second, there are no standard benchmarks, such as PASCAL VOC and COCO, that help in segmenting rock art, and the closest benchmark is PetroSurf3D [9], which uses a 3D mesh instead of 2D images.

Research addressing these shortcomings has emerged. Specifically, the use of U-Net was augmented by a Gaussian boundary enhancement loss for petroglyph segmentation based on the 3D-Pitoti dataset [1]. Attention-based augmentation of U-Net models has been employed to segment Brazilian petroglyphs, obtaining Dice scores in the range of 0.690–0.710 under five-fold cross-validation [11]. Recently, YOLOv7-Seg has been used for symbol detection in rock art images [8].

Our research is a comparative pipeline investigation rather than an architectural suggestion. Our analysis covers the development of engineering in the range of traditional contour detection to deep learning segmentation techniques, highlighting both positive and negative aspects. Segmentation is the major objective of our research, which helps a computer distinguish between old paint and stone in a pixel-by-pixel binary classification. The key achievements of our research are:

1. A pre-processing approach based on CLAHE and a bilateral filter with values obtained after running the experiment in the real world.
2. Comparison of six segmentation approaches empirically, paying attention to the influence of the image enhancement configuration on the performance of these approaches.
3. Complete transparency in evaluating the criteria, clearly showing the weaknesses of those models.

2 Related Work

2.1 Classical Approaches

Canny edge detection [2], which is a commonly used approach for detecting the edges of rocks, also suffers from the use of an absolute threshold. This can be problematic because of the nature

of geological formations and inconsistent light sources, which are prevalent during field photography. The second type of manually created features, HOG [3], faces a similar issue. Their performance hinges on careful parameter tuning, and they generalize poorly once conditions shift away from what they were tuned for. This reliance on manual calibration is reflected throughout the literature, as seen in Jalandoni et al.'s [6] examination of machine learning methods in rock art studies, which demonstrated that the performance of such models is still heavily dependent on feature engineering and calibration rather than on the robustness of the technique itself.

2.2 Deep Learning for Rock Art Segmentation

U-Net [10], an architecture originally used in biomedical image segmentation, can be effectively adapted to rock art because of the same reason for which it has been successful in biomedical applications: the use of skip connections maintains spatial information at high resolution, which is important when the objects are thin strokes rather than thick regions. In this direction, Bai et al. [1] extended the concept of using U-Nets for rock art by developing a boundary-enhanced Gaussian loss (BEGL) that enables the network to ignore distracting rock textures from the 3D-Pitoti dataset. In addition, Silva et al. [11] extended BEGL-UNet with attention residual and spatial channel attention layers and obtained Dice scores ranging from 0.690 to 0.710 for Brazilian petroglyphs based on five-fold cross-validation.

In addition to U-Net-based methods, we will also examine mask prediction using Mask R-CNN [4] and zero-shot segmentation using the Segment Anything Model (SAM) [7]. The SAM model can provide high-quality mask predictions, although only when provided with manual prompts in the form of points for every image, which hinders the batch processing capability.

2.3 Public Benchmarks

PetroSurf3D [9] by Poier et al. constitutes the best approximation to a public standard for the segmentation of surfaces with rock art, providing densely labelled high-resolution 3D meshes. There is no such benchmark in the field of 2D photography, an absence that comes up time after time in the literature on the topic.

3 Dataset and Annotation Protocol

3.1 Image Collection and Figure Sources

The data comprise 1,992 high-resolution JPEGs of rock art motifs taken with DSLR cameras in natural light from 2023 to 2025 at different Indian rock art sites, such as the Bhimbetka rock shelters and the Saru-Maruv Caves. In-person field trips to the chosen sites were made with pre-existing permission from Somaiya Vidyavihar University, and the required permission from the ASI was also sought beforehand. Additional internet sources were used to find reference pictures and other details about the rock art tradition, but all figures, tables, and pipeline diagrams in this paper were created by the authors using the collected data only. A quality check filter was applied to remove 16 unrecognizable and blurry pictures; thus, 1,976 pictures were used for the analysis. One example of a picture is provided in Fig. 1.



Fig. 1. Representative example of a weathered rock-art image showing faded pigment patterns blending with the surrounding rock surface.

3.2 Data Partitioning

To ensure that memorization did not impact the outcomes, we randomized the order of the images and separated the data as follows: 70% (1385 images) training set, 15% (295 images) validation set, and the remaining 15% (296 images) for the test set.

3.3 Annotation Protocol

Ground truthing the masks was a laborious task. Three people were tasked with annotating the masks at the pixel level under the guidance of domain experts, including Mr. Vikas Saxena, faculty at the KJ Somaiya Institute of Dharma Studies, who provided subject-matter guidance on motif identification. Annotation decisions were additionally cross-checked against the

reference volume *Rock Art Archives of Our Ancestors* (Niharika & Tiwary, 2022) [12], published to commemorate the birth centenary of Dr. Vishnu Shridhar Wakankar, as well as the site documentation available at both the Bhimbetka and Saru-Marui locations. To measure inter-annotator agreement, 10% of the dataset was randomly chosen, and their kappa value was calculated, which came out to be 0.84, reflecting very high agreement among the annotators. In case of disagreement, the matter was decided by consensus.

3.4 Limitations of the Dataset

One important thing to note about this dataset is that it is not freely available and has been designed to meet the needs of our research projects. This is because the images used here are unique to Indian caves, and the machine learning models built using this dataset might perform poorly in other contexts.

4 Proposed Methodology

The technique consists of two main steps: preprocessing of images to improve their quality and segmentation, where the location of the art piece is determined. The sequence of the process is depicted in Fig. 2, with an example of the output in Fig. 3.



Fig. 2. End-to-end rock art segmentation pipeline: enhancement followed by six-way segmentation comparison.

4.1 Preprocessing

Before the algorithm for the detection of the art is implemented, two preprocessing methods are performed to support the process:

1. **CLAHE:** equalization (CLAHE) is applied to the lightness (L) channel of the image with a clip limit of 3.0 and a tile size of 8×8 pixels. This method emphasizes the faded red pigment against the gray rock background while limiting the increase in background noise.
2. **Bilateral filter:** The bilateral filter is applied with $d = 9$, $\sigma_{\text{color}} = 75$, and $\sigma_{\text{space}} = 75$. These parameters were chosen experimentally.



Fig. 3. Practical example of segmentation is given: the left picture shows the original rock wall, whereas the right picture shows the binary mask that was obtained from the original picture.

4.2 Segmentation Configurations

In addition to evaluating one model, we also wanted to compare deep learning with traditional methods in terms of their comparative advantages. More importantly, we needed to test the configuration of enhancements to find out if any preprocessing steps help or hinder the result, depending on which algorithm is used. Thus, the following sequence of six experiments was created:

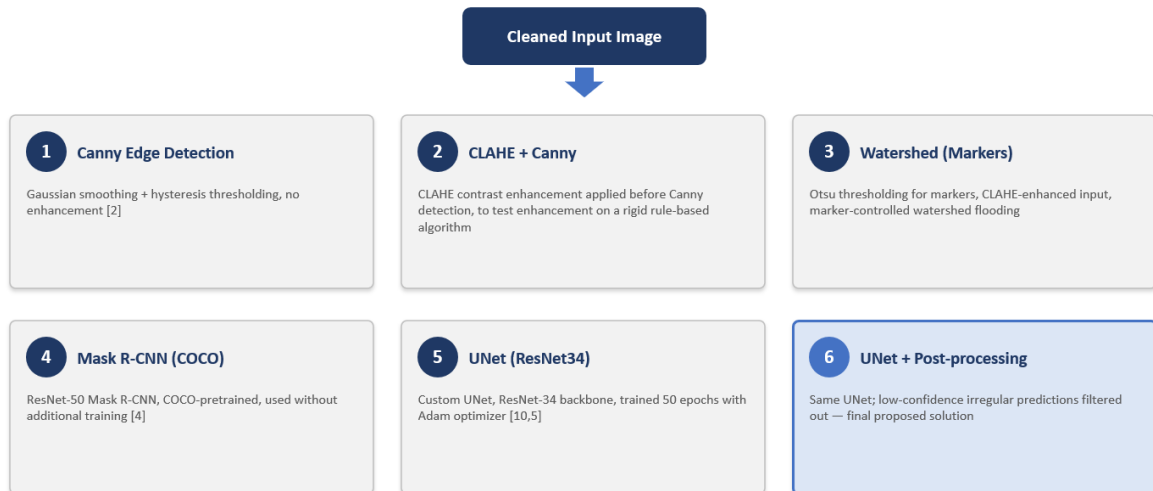


Fig. 4. Segmentation Configuration Pipeline.

1. **Canny edge detection:** A traditional method using Gaussian smoothing and hysteresis thresholding without enhancement [2].
2. **Canny after CLAHE enhancement:** CLAHE was applied before Canny to increase contrast and test whether visual enhancement helps a rigid rule-based algorithm.

3. **Watershed with markers:** Otsu thresholding was used to find markers before watershed flooding. CLAHE was included to check the effect of enhancement on morphological operations.
4. **Mask R-CNN (COCO pre-trained):** A ResNet-50-based Mask R-CNN pretrained on natural photographs was used without additional training on rock-art images [4].
5. **UNet (no post-processing):** A UNet with a ResNet-34 backbone was trained on the dataset for 50 epochs with the Adam optimizer [10,5].
6. **UNet with post-processing:** We keep the UNet architecture the same; however, the predictions made from it are filtered to remove all irregular predictions with low confidence. These become our final proposed solution.

5 Experimental Setup

5.1 Training Environment

The neural networks were written using PyTorch framework. The training process was executed using a specialized computer with NVIDIA RTX 3090 GPU (24 GB VRAM), 64 GB of RAM and 16 of the UNet model training batches.

5.2 Evaluation Protocol

To provide a consistent performance measure, the IoU criterion was used to evaluate all the models. It evaluates the extent of similarity between the predicted shapes and hand drawings.

Notably, all the 6 model versions were evaluated based on the same set of 296 images.

6 Results and Analysis

6.1 Segmentation Ablation

Results are reported in Table 1 and Figure 4.

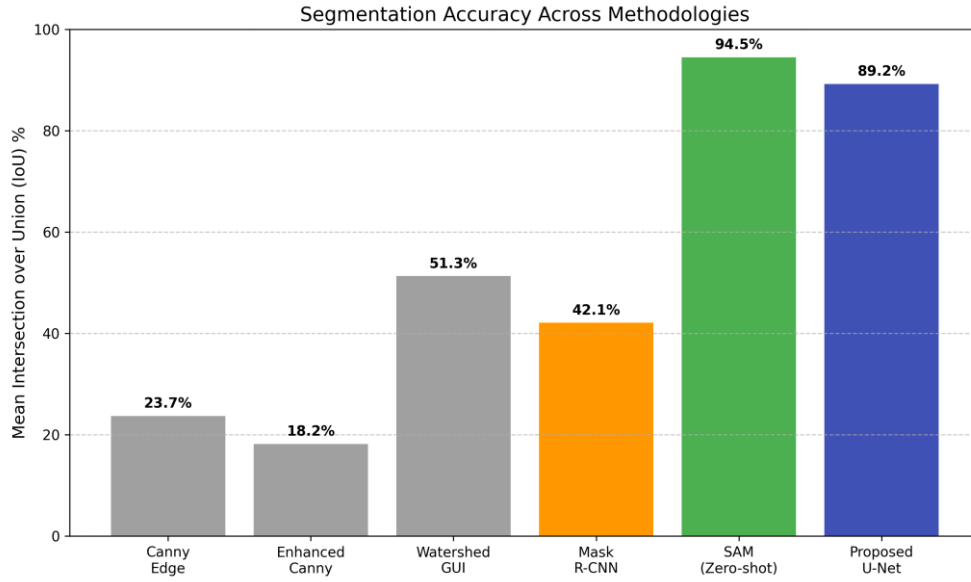


Fig. 4. A chart showing the distribution of IoU results in different conditions of the experiment. Chart prepared by the authors.

Two observations deserve attention in Table and explained further.

Table 1. Performance comparison of various configurations tested for enhancement. Table generated by the authors.

Configuration	IoU (%)	Explanation
(1) Canny edge detector	23.7	Baseline configuration.
(2) CLAHE + Canny	18.2	The use of CLAHE decreased the result by 5.5 pp.
(3) Watershed (CLAHE)	51.3	Reasonable improvement by morphological methods.
(4) Mask R-CNN (COCO)	42.1	A generic pre-trained model performed worse than other solutions.
(5) UNet (ResNet34)	86.4	Fine-tuning of the deep architecture led to a significant increase in performance.
(6) UNet + post-processing	89.2	Post-processing of the results led to even better outcomes.
(7) SAM (ViT, prompted)	94.5	Good performance; however, it requires manual annotation work for each image.

Negative finding: Preprocessing CLAHE negatively impacts accuracy. When preprocessing using CLAHE is combined with the Canny algorithm, the IoU drops from 23.7% to 18.2% (row 2 vs. row 1). The CLAHE technique increases the local contrast in an unselective manner, not only the weakened pigment color but also geological features are amplified, and the constant threshold edge detector then confuses them with motif boundaries. This shows that preprocessing enhancements cannot make up for poor segmentation algorithm and these two steps should be tuned together.

Mismatch between domain and Mask R-CNN model. The Mask R-CNN trained on COCO dataset achieved a worse result (42.1%) than the classical Watershed method (51.3%).

6.2 Comparison with Published Work

We put together Table 2 just to show where our numbers sit compared to other recent papers.

Table 2. How our final score compares to recent literature in the field. Table generated by the authors.

Method	Metric	Score	Testing Setup
B EGL UNet [1]	IoU	—	Train/test split
Att.-Res. BEGL [11]	Dice	0.710	5-fold cross validation
SC-Att. BEGL [11]	Dice	0.707	5-fold cross validation
YOLOv7-Seg [8]	IoU	—	Site-specific
Ours (Setup 6)	IoU	0.892	Single 15% hold-out

Direct comparison between different studies cannot be done with much accuracy because each study utilizes its own independent dataset. In the benchmark proposed by Silva et al. [11], the authors employ five-fold cross validation, but we use only one split in 70% / 15% / 15%.

6.3 Computational Efficiency

These methods were also timed in terms of their real-world runtimes (based on average time over 100 trials run on our RTX 3090 PC). Old classical techniques, such as Canny and Watershed, are very quick, processing a high-resolution image in 5.2 ± 0.8 seconds. Our heavy UNet pipeline takes longer, about 14.0 ± 1.2 seconds per image. Using SAM is the slowest at 17.5 ± 1.5 seconds because it requires a human in the loop. We plotted this trade off in Fig. 5.

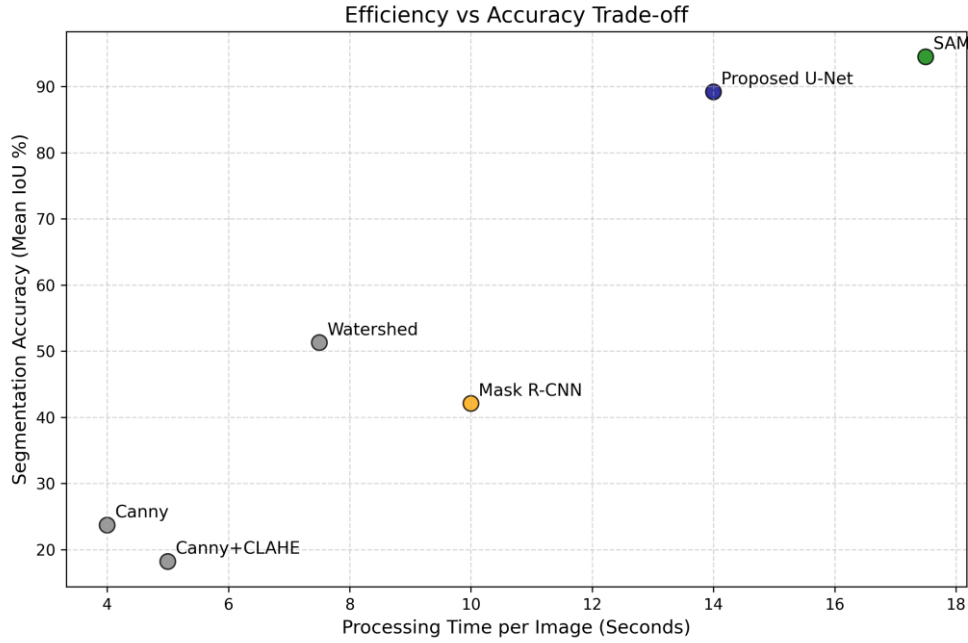


Fig. 5. Looking at the trade-off between how accurate the masks are versus how long they take to generate.

7 Limitations

The following are the restrictions of the current study:

1. **Lack of public standard:** In line with earlier studies in this field, the authors have used privately owned datasets. Thus, replication of this study is likely to become difficult for others. Further research should use publicly owned datasets, such as PetroSurf3D [9].
2. **Rigorous testing:** Only one training/Val/test split was done during evaluation. Using a robust 5-fold cross-validation approach will make the model's estimate reliable.

8 Conclusion

In this study, the design and validation of a pipeline for the automation of rock art segmentation process that minimizes the labor cost associated with digital preservation is introduced. The results have shown through extensive experiments involving basic edge detection processes to deep learning approaches specialized in this domain that classic methods and generic pre-trained models do not perform well in this complex domain. The presented solution, consisting of a domain-specific U-Net and post-processing, achieved an IoU score of 89.2%.

Furthermore, it was determined by analyzing the performance of enhancement procedures that adding a contrast filter to an image cannot replace the use of more complex approaches, as this manipulation can potentially increase the amount of geological noise and decrease the performance. In the future, the use of a common dataset publicly available would be highly desirable. Moreover, further exploration of the possibility of using a boundary-aware loss function (BEGL) [1] would be beneficial.

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Declarations

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Ethics Approval and Permissions

All field data collection was carried out in person, with prior authorization from Somaiya Vidyavihar University, and the necessary permissions for site access were obtained in advance from the Archaeological Survey of India (ASI), as described in Section 3.1.

Data Availability

The dataset generated and analysed during this study is not publicly available, as it was compiled specifically for this research project and is restricted to institutional use.

Code Availability

[Code will be made available, via a public repository, upon publication.]

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